Abhinav Gyawali

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Research Interests

Database systems, incremental computation, unstructured data analysis and fault-tolerant distributed systems.

Education

University of Texas at Austin — Aug 2026 – Present
PhD in Computer Science
Advisors: Dixin Tang, Vijay Chidambaram

Tribhuvan University — Oct 2019 – Aug 2024
BS in Computer Science and Information Technology

Publications

Mihai Budiu, Leonid Ryzhyk, Gerd Zellweger, Ben Pfaff, Lalith Suresh, Simon Kassing, **Abhinav Gyawali**, Matei Budiu, Tej Chajed, Frank McSherry, Val Tannen. — VLDBJ 2025

DBSP: Automatic Incremental View Maintenance for Rich Query Language.
The VLDB Journal, 2025. (The "Best of VLDB 2023" special issue.)

Research & Industry Experience

Feldera | Software Engineer — US, Remote — Dec 2023 – July 2026

- Designed and implemented a distributed, fault-tolerant pipeline execution system with S3-backed checkpointing, crash recovery, and real-time output connectors for **Postgres** and **Redis**.
- Extended Feldera's SQL compiler with support for complex types (**VARIANT**); built automated differential testing infrastructure using **SQLancer**, uncovering **31 previously undetected bugs**.
- Developed the **Feldera Python SDK** for programmatic pipeline control and authored technical documentation covering stream processing, batch processing, and observability.

Invisid | Software Developer — Germany, Remote — May 2022 – Nov 2023

- Developed **trackpad-rs**, a Rust library for high-fidelity trackpad and mouse interaction data from Apple's *MultitouchSupport* framework, enabling continuous biometric authentication.
- Redesigned backend server infrastructure for behavioral authentication, improving efficiency, scalability, and consistency of biometric data processing.

Flow Webinar | Security Engineer (Part-time) — US, Remote — May 2020 – June 2021

- Conducted vulnerability assessment and penetration testing of a WebRTC-based video conferencing platform; managed backend infrastructure and system administration for production deployments.

Selected Projects

DBSP | Rust — feldera/dbsp

- Incremental computation engine where change propagation runs in time proportional to the change size rather than the full dataset – the engine underlying the VLDBJ publication.

SocketDB | Rust — abhizer/socketdb

- Lightweight SQL database that surfaces real-time query updates to clients over WebSockets.

Nyx-lang | Rust — abhizer/nyx-lang

- Statically typed, tree-walking interpreted language with a full type-checking pass for safety and correctness.

Loogle-rs | Rust — abhizer/loogle-rs

- Local full-text search engine based on TF-IDF ranking.

Honors & Awards

UTCS Distinguished Graduate Fellowship | University of Texas at Austin — 2026

SIGPLAN-M Mentee | Mentor: Prof. Jonathan Balkind — 2025

DWIT Merit-Based Scholarship | Deerwalk Institute of Technology — 2020, 2021